

Interpersonal Trust Perceptions in a Hybrid Space: Understanding Social Dynamics in Location-Based Games

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Abstract

Location-based games (LBGs) create hybrid spaces where digital and physical worlds converge, enabling new social interactions but leaving under-explored social dynamics regarding interpersonal trust. This study investigates interpersonal trust perceptions through in-depth interviews with 26 participants from four LBGs: Pokémon GO, Monster Hunter Now, Ingress, and Pikmin Bloom. Findings highlight the critical role of online communities (e.g., Discord, Facebook) in fostering initial trust for real-world meet-ups through the “online to offline” and “group to individual” scaffolding, alongside the careful risk-benefit calculations players make when sharing location data through game interactions. Based on these insights, I propose design recommendations, such as integrating seamless online community entry points and improving location privacy controls within games. This work contributes to an empirical understanding of interpersonal trust perceptions in LBGs and offers theoretical insights into how digital and physical interactions facilitate new forms of social collaboration within hybrid spaces.

CCS Concepts

• Human-centered computing → Empirical studies in HCI.

Keywords

Location-based Game, Interpersonal Trust, Interview, Empirical Understanding

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1 Introduction

Location-based games (LBGs) combine physical and digital spaces, overlaying real-world environments with interactive virtual elements that players engage with as they move through their physical surroundings. Unlike traditional video games, LBGs stimulate real-world exploration, fostering social interaction and physical activity, turning ordinary routines into playful experiences. One

of the most successful examples is *Pokémon GO* [31], which has cultivated a global player base and demonstrated benefits such as enhancing players’ well-being, fostering social connections, and promoting outdoor engagement [41, 43]. Alongside *Pokémon GO*, other LBGs, such as *Pikmin Bloom* [28], *Ingress* [29], and *Monster Hunter Now* [30], also flourished and have embraced popularity in recent years.

In LBGs, *Interpersonal Trust* [35, 39] refers to a player’s confidence that another player or non-player will act in a way that fulfills their expectations and avoids harm, which is central to social interactions and gaming experience. While engaging with unfamiliar individuals is common in LBGs [41, 46], prior research on LBGs highlights players’ concerns regarding interpersonal trust, such as sharing location data [5] and face-to-face communication with unknown individuals [4, 24]. Despite rising awareness of the interpersonal trust concerns among LBG players [12, 21], how players build interpersonal trust and navigate these concerns remains unclear, highlighting a need for further research on interpersonal trust in LBGs.

Gaining an empirical understanding of interpersonal trust perceptions among LBG players is crucial. LBGs represent a unique and emerging *hybrid space* [11] social context, blending online and offline interactions that differ from traditional settings. Trust is a frequently discussed topic in the HCI research community, but simulating trust situations in a laboratory setting is challenging [35]. In this context, LBG players’ lived experiences offer vivid data for investigating interpersonal trust perceptions within hybrid spaces. An empirical understanding not only enables the development of informed design recommendations to enhance the LBG experience but also contributes to the broader HCI field by offering valuable insights into trust dynamics and interaction design in non-gaming contexts, such as location-based services and gig work platforms. In this work, I conducted a qualitative study using in-depth interviews with 26 players from four major LBG communities: *Pokémon GO*, *Monster Hunter Now*, *Ingress*, and *Pikmin Bloom* (See Figure 1).

Through an exploratory research approach [40] and thematic analysis [2], I investigated players’ perceptions of interpersonal trust, and this work is guided by the following research questions: **RQ1**: How do LBG players develop interpersonal trust with unfamiliar players? **RQ2**: How do LBG players navigate and manage risks regarding interpersonal trust? In sum, this work contributes:

- an empirical understanding of interpersonal trust in LBGs, providing design implications for more trustworthy and engaging LBG experiences.
- Identify the unique scaffolding trust-building dynamic from online to offline, from group to individual in the context of

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Figure 1: User interface for (a) *Pokémon GO*, (b) *Ingress*, (c) *Monster Hunter Now*, (d) *Pikmin Bloom*.

LBG, which extends the literature and provides theoretical insights into trust perceptions in hybrid spaces.

2 Background

2.1 Interpersonal Trust in Digital Games

Interpersonal trust is an individual’s subjective perception of others’ intentions [44] and encompasses multi-dimensional, dynamic definitions that vary across contexts [1, 33, 44]. Drawing on definitions from Deutsch [14], Mayer et al. [26], Rothstein and Stolle [34], and Abdul-Rahman and Hailes [1], interpersonal trust in LBGs can be defined as a player’s confidence that another player, or non-player encountered during the LBG experience, will act in a way that fulfills expectations and avoids harm. Rather than representing a binary choice of “trust or distrust,” interpersonal trust in LBGs operates as a spectrum. In other words, players weigh their desire to engage with the game—often involving interactions with unfamiliar individuals—against the perceived risks of potential harm from those interactions.

Multi-player games have been shown to foster interpersonal trust among participants. Depping et al. [13] highlight that social games often act as icebreakers, enabling trust-building through shared activities. Similarly, Chen et al. [8] examine how digital games foster generalized trust [39], showing that shared gaming experiences can transform strangers into collaborators and cultivate trust through joint play. Specifically, LBGs have been shown to serve as social platforms that foster connections among strangers [12, 16–18].

2.2 LBGs as Hybrid Spaces

LBGs present unique challenges to trust-building due to their nature as hybrid spaces, where digital and physical interactions converge. Unlike traditional multiplayer digital games, which occur primarily in virtual environments, LBGs require players to navigate a blend of online coordination (e.g., through communities on platforms like Discord) and chance in-person meetups [4]. This integration of online and offline interactions, combined with the use of real-world locations, introduces new layers of complexity to interpersonal trust dynamics. Researchers have argued that hybrid spaces challenge conventional understandings of trust and safety in society, redefining how people interact and collaborate in these interconnected environments [11].

Meanwhile, prior works highlight various trust concerns in LBGs. Bhattacharya et al. [4] analyzed group interactions in *Pokémon GO* and identified challenges such as meeting unfamiliar players in person, fears of theft, traveling to unfamiliar or unsafe locations, traffic accidents, and risks associated with being out after dark. Similarly, Blasiola et al. [5], in their study on *Ingress*, emphasized concerns such as location privacy and potential disruptions for non-players. Broader studies on LBGs and location-based services (LBS) echo these concerns, highlighting issues related to privacy, security, and ethical considerations [10, 15, 20, 24, 25, 32, 36]. While existing research provides valuable insights, much of it either examines interpersonal trust within broader contexts, limiting its depth of empirical understanding or takes a deductive approach that primarily serves as a cautionary framework [3]. Prior research in Couchsurfing services [47] has shown that online platforms with user profiles and communication features can facilitate online familiarity, which in turn supports the development of offline trust. Building on this body of work, my study aims to delve deeper into how LBG players build interpersonal trust with unfamiliar players and navigate and manage interpersonal trust concerns.

2.3 Contextualizing the LBGs

In this work, I chose four major LBGs as research probes [19]. *Pokémon GO* [31] is a location-based adaptation of a globally popular intellectual property (IP) [45]. It encourages players to explore their surroundings to capture Pokémon, promoting social engagement through multiplayer activities like raids. Raids involve challenging battles requiring co-located teamwork and fostering community among players. The core gameplay loop includes catching, evolving, and battling Pokémon in gyms and raids for rewards. *Ingress* [29] transforms the world into a strategic game where players join factions (Enlightened or Resistance) to capture and link real-world portals. The game fosters identity and teamwork as players gather resources, establish control fields, and compete for dominance. *Monster Hunter Now* [30] integrates Capcom’s *Monster Hunter* IP with real-world exploration. Players collaborate to defeat massive Wyverns, collect materials, and craft gear to take on even tougher foes. The core loop involves tracking Wyverns to specific locations and battling them until a winner emerges. *Pikmin Bloom* [28] reimagines Nintendo’s *Pikmin* series as an LBG. Players explore, discover Pikmin, and beautify the world by planting flowers. The gameplay loop centers on walking, growing Pikmin, and participating in community events to create blooming landscapes.

3 Methodology

To address the proposed research questions, I adopted a qualitative research approach to gain insights into LBG players’ perceptions of interpersonal trust. I conducted in-depth semi-structured interviews with 26 participants and applied inductive thematic analysis [2] to analyze the interview data.

3.1 Participants

After receiving approval from the Institutional Review Board (IRB), I recruited 26 participants directly for one-on-one interviews conducted between March and April 2024. Recruitment was carried out through social media platforms, such as Reddit and Instagram.

Interested individuals were asked to fill in a prescreening survey to ensure they are above 18 years old and have at least one LBG playing experience. The interviews were scheduled based on participants' availability, and each session lasted approximately one hour. Upon completion of the interview, participants received a 30 USD digital Amazon gift card as compensation. Recruitment continued until theoretical saturation was reached [7]. The 26 participants—10 females, 14 males, and 2 non-binary people ranging in age from 22 to 67 and residing in six different countries across multiple continents—completed the interviews. The 26 participants' experiences encompass four LBGs: *Pokémon GO* (22 players), *Ingress* (7 players), *Monster Hunter Now* (5 players), and *Pikmin Bloom* (5 players). Among them, 10 participants engaged with multiple LBGs. All participants played LBGs for at least one year, and 20 of 26 engaged with LBGs daily. It is worth noting that this sample may introduce bias, as the majority of participants are from the *Pokémon GO* community, which reflects the general player base of LBGs. To ensure confidentiality, all participant names are pseudonyms.

3.2 Interviews

All interviews were conducted one-on-one online via Zoom sessions, and the audio from each session was recorded for transcription purposes. Each interview lasted approximately 50 to 60 minutes. To ensure the interviews effectively explore LBG players' perceptions of interpersonal trust rather than focusing on specific game features, the semi-structured interview questions are carefully drafted and refined with my supervisor. Participants were encouraged to elaborate on their responses with follow-up questions such as, "Why" to clarify their concerns and "How" or "What" to provide specific examples. For instance, if a participant expressed concern about meeting a player in person for the first time, I probed further with questions like "Why does this worry you?" In addition, I asked participants to share any opinions, experiences, and beliefs they feel are related to interpersonal trust in LBGs. As such, semi-structured interviews allowed our participants to share their experiences and perceptions thoroughly [6]. Meanwhile, detailed notes were taken for each interview to support the analysis.

3.3 Data Analysis

After each interview, a preliminary analysis of the data was carried out to identify potential themes, which were then critically discussed with my supervisor. This iterative process continued until no new themes emerged from the interviews, at which point participant recruitment could be halted. The audio recordings of the 26 interviews were initially transcribed automatically by Zoom and then manually corrected. During this process, I anonymized place names and used pseudonyms for the names of the participants. These transcripts and notes were uploaded to the professional qualitative analysis software MAXQDA 24¹. I used inductive thematic analysis [2]. This involved three primary stages: 1) Familiarization with the data, which included reading the transcripts, making corrections, and annotating with memos; 2) Generating initial codes by identifying phrases and sentences pertinent to our research aims; 3) Constructing themes by aggregating all relevant codes. The first author developed the initial codes and themes, which the

research team then reviewed and refined collaboratively. The finalized themes were articulated as titles in the Findings section, encapsulating participants' interpersonal trust perceptions in LBGs.

4 Findings

4.1 Scaffolding Trust Building From Online Communities to Offline Meet-ups

4.1.1 Online Vetting as a Pre-step. The collocated multi-player mechanism in LBGs requires players to gather in groups to complete specific tasks in the game (e.g., complete EX raid battles in *Pokémon GO*). All participants mentioned they coordinated meet-ups in online communities (e.g., Discord, Facebook) in advance and gathered at planned events; ad-hoc meet-ups without pre-planning are relatively rare nowadays. Some participants (7/26) mentioned due to safety concerns, they only chose to play with friends and families and avoided playing with strangers, though they recognized this decision would make them miss many group events and rewards in games.

In contrast, the rest of the participants, many of them (19/26), in order to make more friends, complete game tasks, or increase their chances of winning battles, played with unfamiliar players they had not previously met in the real world. However, they would take cautious consideration beforehand. For instance, Alice mentioned that she had to have ice-breaking conversations before their real-world meet-ups: *"I am wary of strangers. If someone I had not played with before invited me to play together, I'm like, oh, hey, if we want to meet up in real life, we at least talk a bit online first. We have to introduce ourselves. I at least have to know you are not weird or dangerous people. If we became online friends before meeting in real life is the best situation. I feel more comfortable meeting them."* (Alice). When asked how much online vetting would help build interpersonal trust with strangers, participants believed that while it might not lead to *"a leap of faith"* (Will), it does play an essential role in boosting confidence when deciding to meet them.

Sometimes, participants may not have engaged in prior communication within the group chat and instead meet by chance in real life. In this situation, participants exhibit heightened caution with meeting unfamiliar players in real life. Kate shared her experience of unexpectedly encountering unfamiliar players when playing *Pokémon GO*: *"I just moved here this month. Last week, I came across a group gathering and playing Pokémon GO. I was a lot more cautious as I knew no one here: I sat nearby and waited and watched to see who turned up, and then if I didn't like the looks of them I might have just not joined them. Then, once I was sure they didn't seem unsafe, I approached them and said, hey, are you guys playing Pokémon GO? That was the case for the first time I met the [City Name] group."* (Kate).

When meeting and playing with unfamiliar players, participants relied on *"common sense to judge the situation"* (Mike). In on-the-spot scenarios, such as Kate's, players often felt uncertain due to the lack of prior knowledge about unfamiliar individuals. However, online communities like Discord and Facebook played a pivotal role in bridging this gap by offering a *"trust scaffolding"* mechanism. These platforms provided opportunities for players to connect and vet unfamiliar players beforehand, enabling more informed judgments

¹<https://www.maxqda.com/new-maxqda-24>

and reducing anxiety during in-person interactions. While these online spaces may not immediately foster a “leap of faith,” they serve as critical environments for initial confidence-building, coordination, and establishing personal comfort and safety boundaries. This scaffolding approach highlights the layered nature of trust-building, where virtual interactions set the stage for smoother transitions into offline engagements. Notably, participants who lacked access to these online communities reported feeling significantly more nervous and uncertain during real-world meetups.

4.1.2 Group Size, Timing, and Location Matter. To many participants (15/26), the size of meet-up groups matters. They feel safer in larger groups than in one-on-one meet-ups involving unfamiliar players. For instance, Gina said: *“I just say no to all one-on-one invitations from people I never met before. Unless I have a group of people with me, 6 or 7 would be the best.”* (Gina). Additionally, most participants (23/26) mentioned they tend to plan meet-ups containing unfamiliar players in public places during the daytime only: *“If I play with friends, time and place do not matter; I am okay meeting them late at night, maybe at someone’s home or workplace. But if it is a gathering with people I have never met before, to make everyone feel comfortable, I would suggest meeting and playing at a public place, more crowded, and daytime only.”* (Sarah). In general, participants tend to trust larger groups over one-on-one meet ups when interacting with unfamiliar players and prefer to meet in public places during the daytime, which are commonly perceived as safer.

4.1.3 Being Recognized as Suspicious by Non-Players. Besides interacting with other players, playing LBGs often leads to encounters with non-players in shared public spaces, and these individuals may not recognize the behaviors of players, leading to concerns around trust and privacy. Kelly shared her experience: *“There was a group event. We all came together, and someone, maybe living around, called the police because they didn’t know why 20 people all clamored into their neighborhoods.”* (Kelly). Such instances highlight how the presence of unfamiliar players in public spaces can provoke suspicion.

In addition to “suspicious group gatherings,” LBGs like *Pokémon GO* and *Monster Hunter Now* incorporate augmented reality (AR) cameras that allow players to capture AR elements in real-world settings. Joan shared her experience with this feature, describing how taking AR photos often generates confusion and unease among non-players: *“When I use AR cameras, I am holding the phone, looking at this or looking at that. You know these are not places for tourists, so it can look weird. People would look at me a little funny and suspicious. Some people would ask: Did you include my face or my child’s face? What are these pictures for? So I’ll just show them the screen and say, I’m just playing a game. You never know if somebody is gonna be totally upset with me or get road rage, so I feel better being careful. Now, I only take AR photos in nature.”* (Joan).

As Joan notes, the act of taking AR photos in LBGs can appear suspicious or intrusive to non-players, raising privacy concerns. Her experience illustrates how non-players are often unaware of the context in which players are engaging, which can easily lead to misinterpretation of players’ intentions. This lack of understanding contributes to trust tensions, as non-players may perceive players’

actions as intrusive or potentially harmful, while players feel the need to be cautious to avoid conflict.

4.2 Concerns About Location Traces

4.2.1 The Worry of Sensitive Locations Being Deduced by Unfamiliar Players. In addition to the tracking done by the app itself, LBG players’ movements on game maps can leave “traces” (Anna) and be seen by other players through location-tagged item exchange and active logs. This design in LBGs intentionally desired to encourage players to explore more locations and create social bonding, but it raised privacy concerns among the participants. Most participants (24/26) self-reported regularly and frequently visit landmarks in LBG maps around their real-world living and working places, which are regarded as “sensitive locations” (Nathan). Traces in the shared digital map could potentially lead others to infer their daily routine and identify their sensitive locations.

Simon shared their concern that other players, especially unfamiliar ones, would figure out their workplace by noticing their frequent presence in that location: *“It is always in the back of my mind. I have a stop in Pokémon GO, and it is reachable from my desk at work. I will, very frequently, send everybody gifts from that stop. Multiple days in a row, I think, wow, if they ever find this stop, they’ll figure out pretty quick this is where I work because it’s the only office building in our area. It kinda scared me. Especially some friends in the game, I do not know them very well in real life.”* (Simon).

Reflecting on Simon’s concern, Emily highlighted an instance in her local group chat where one player tried to guess another player’s home address based on their frequent activity on the game map: *“Someone in our group chat put it through other players’ usernames and tried to basically tell everyone, hey, guess where this person lives, and then try to ask people to approve that. Obviously, people aren’t gonna respond anything to that, and this person was kicked out, but the fact that they’ve put another person’s home address is too scary.”* (Emily). Given that others can easily infer players’ sensitive locations, participants expressed significant concerns and emphasized the need to protect their sensitive locations and movement patterns from being deduced.

To address this issue, participants who play *Pikmin Bloom* appreciated the game’s “private zone” feature, which allows players to designate customizable areas on the map where their movements are automatically hidden from others. However, this privacy feature is not a common practice across all LBGs.

4.2.2 Spontaneous Protective Strategies. In response to this concern, only a few participants established their own strategies to circumvent these concerns. Anna, for example, explained that she would not play at the same places at the same time every day. Her strategy was to *“consciously avoid a routine so it is impossible (for others) to understand my movement patterns.”* (Anna). Likewise, Rita often sent location-tagged game items away from her sensitive locations, or if she really wanted to send out items around her home, she would only send them to people she trusted or international players who were *“too far away, so no threat.”* (Rita). Most participants, despite concerns that others with malicious intent might compromise their location privacy, tried to downplay these worries to focus on enjoying the game. They were either optimistic like Simon: *“The risk is always there, but so far, nothing bad has happened*

to me.” (Simon). Or they felt unsure what can be done to mitigate concerns: “I know it is a risk-benefit calculation, but I don’t know how we can share these in-game items in a safe way.” (Julia).

Interestingly, participants are aware that sharing their location in a game carries privacy risks, but they prioritize social interaction and the enjoyment of connecting with other players. While they understand that some players might not respect privacy boundaries, they tend to ignore or downplay the potential dangers of encountering individuals with harmful intentions. Essentially, the desire for social engagement outweighs their concerns about privacy, even though they recognize the risks involved.

5 Discussion and Conclusion

This study is the first to explore interpersonal trust perceptions among LBG players. Through interviews with 26 participants, I revealed three key aspects: how online communities help build trust with unfamiliar players, players’ concerns about location privacy, and trust tensions between players and non-players. In this section, I discuss these findings, provide design recommendations, and reflect on the study’s limitations and future research directions.

5.1 Scaffolding Building Approach for Interpersonal Trust: Online to Offline, Group to Individual

The findings underscore the critical role of online communities in fostering interpersonal trust for collocated play with unfamiliar players. By connecting individuals through virtual platforms, these communities create a structured foundation for confidence-building and effective coordination before in-person interactions. This scaffolding approach reaffirmed prior research on trust-building from online to offline [42, 47] and extends it from a group interaction aspect. Beyond the online-to-offline scaffolding, online communities create group environments where interpersonal trust is anchored in institutional trust [37]. Institutional trust refers to the confidence placed in the structures, rules, and norms fostered in online communities.

Within these online communities, players tend to rely on the shared credibility, accountability, and collective endorsement of the group, which fosters a sense of safety and reliability. This foundation of institutional trust allows players to feel secure in engaging with the group, ultimately enabling the development of interpersonal trust during individual interactions. However, the absence of integrated online communities directly within the games, such as Discord, Facebook, or Niantic’s Campfire [27] as external apps, limits the accessibility and effectiveness of this trust-building. Players who do not actively seek out external tools may miss out on these valuable scaffolding opportunities.

Design Implication 1: LBGs should integrate clear and accessible entry points to online communities directly within the game interface, tailored to players’ locations. This integration would enable players to easily connect with others, fostering interpersonal trust in online spaces before transitioning to in-person meetups. By reducing barriers to interaction, players could participate in physical gatherings with greater confidence and a stronger sense of security.

5.2 Concerns About Location Privacy, But Downplayed It

Our findings reveal a nuanced risk-benefit calculation when players face location privacy concerns. While participants expressed worries about the potential misuse of their sensitive location data by individuals with malicious intent, most players tended to downplay these concerns, with a belief that the worst-case scenarios were unlikely to occur.

On the one hand, this behavior can be attributed to generalized trust [39], where individuals exhibit an inherent belief in the goodwill of others. This tendency is particularly pronounced in LBGs, where mistrusting others could lead to missing out on collaborative gameplay opportunities that often rely on some degree of location sharing. Aligning with trust literature [35], people are more likely to trust others when the potential consequences of not trusting them outweigh the risks of doing so. In the context of LBGs, the perceived cost of missing social interactions and gameplay significantly outweighs the potential risk of having sensitive location information inferred. On the other hand, this dynamic can also be explained by institutional trust [37]. Players may perceive the LBG community as a shared social group, fostering a sense of collective identity. This trust in the community can be transferred to individual members, leading players to believe that malicious actions are unlikely within their shared ecosystem. Despite this tendency to trust, concerns about location privacy remain valid. Reports of bad actors and incidents within LBGs highlight the limitations of relying solely on generalized or institutional trust. The current lack of effective safeguards leaves players vulnerable and reliant on trust alone to navigate these risks.

Design Implication 2: LBGs should implement features to empower players in managing their location privacy. For instance, games could provide reminders alerting players when specific actions—such as location sharing or sending location-tagged items from repeated locations—might disclose sensitive information. Additionally, offering options to anonymize certain activities or customize the visibility of shared data would give players greater control over their privacy and reduce their reliance on generalized or institutional trust.

5.3 Trust Perceptions in Hybrid Space

This study offers theoretical insights into trust dynamics in LBGs as hybrid spaces where digital and physical interactions intersect. Beyond online-to-offline platforms like Couchsurfing [47], which uses online platforms to build familiarity and foster offline trust, LBGs scaffold trust-building across virtual group coordination and real-world engagement. In addition, LBGs go further by introducing location-specific interactions and collaborative gameplay that blend virtual and physical spaces. This extends prior research on trust in virtual [9, 23] and offline spaces [22], revealing a layered, iterative trust-building and risk-benefit calculation process.

5.4 Limitations and Future Work

Our study has several limitations. First, participants were all experienced players, which may have influenced their trust perceptions due to their familiarity with the community and game. Second,

while qualitative methods provided in-depth insights, incorporating quantitative methods like surveys could validate and generalize these findings. Third, our study did not account for demographic factors such as gender or age, which could affect trust and safety perceptions, as suggested by previous research [38]. Future studies should consider these factors to better understand interpersonal trust dynamics in LBGs.

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